

Public Investment Evidence

Unstructured reviews for public-project analysis

CIS 509 Team 4

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The goal: a better-informed project proposal

“If we build a project here, what local activity might follow per dollar?”

Completed projects

Inspect outcomes, review language, coverage and source assumptions.

Proposed projects

Enter local conditions, compare cases and examine an experimental estimate.

Large text coverage, small prediction sample

1.06M

unique reviews scored

11

projects across 7 cities

10

eligible project outcomes

Learned-topic analysis uses 23,691 sampled reviews

10,971 baseline reviews fit the descriptive vocabulary and topics.
The archive describes historical business-review activity.

Review text becomes place-specific evidence

“The only unfavorable part was the sidewalk seating since the city hadn’t picked up the garbage.”

Topic

**Cleanliness /
maintenance**

Target

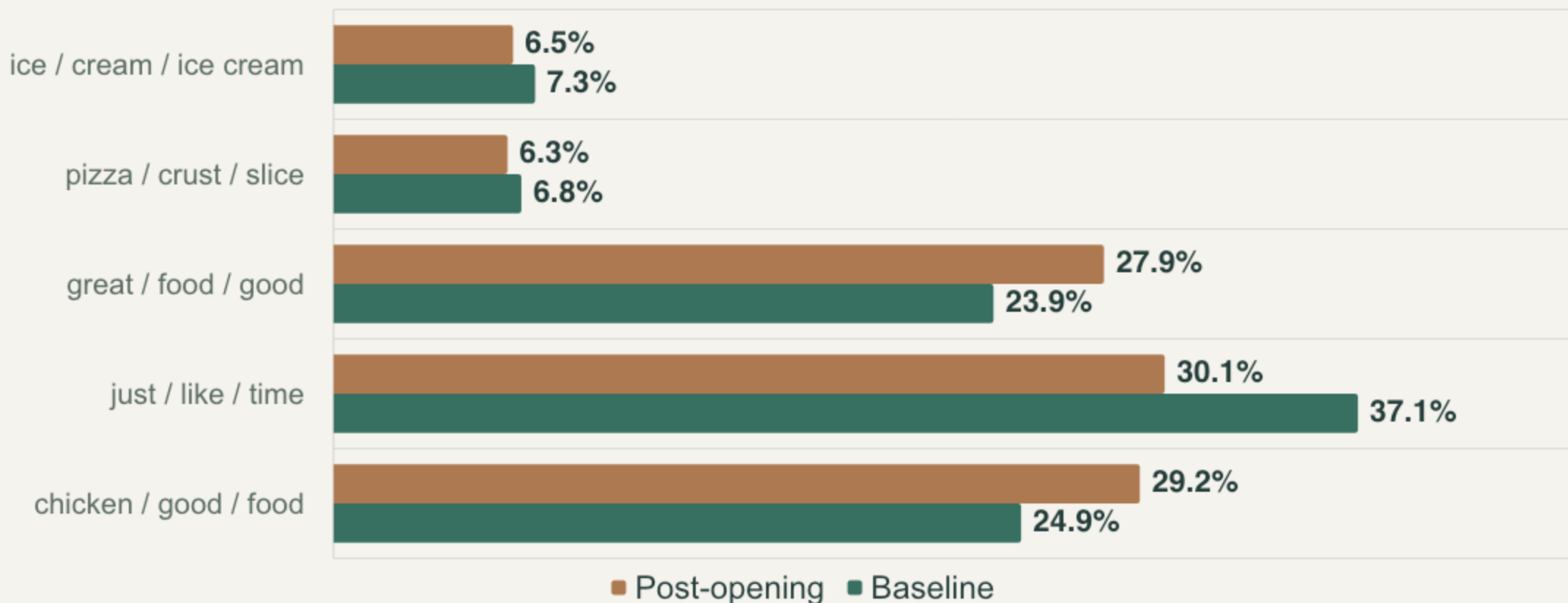
Surrounding area

Clause polarity

Negative

Learned topics expose what reviews discuss

TF-IDF unigrams and bigrams with five NMF topics



Dilworth Park: 1,200 sampled reviews per period. Mean normalized topic weight.

Raw growth can trail the comparison area

Sun Link review counts	Reviews
Nearby baseline, 2010–2011	3,807
Nearby post, 2015–2016	10,943
Expected post from comparison growth	13,150.3

–11.2

adjusted reviews
per \$1M

$$\text{CE} = (\text{observed nearby post} - \text{expected nearby post}) \div \text{cost in \$M}$$

Descriptive comparison. Other neighborhood changes can influence review activity.

Advanced text did not improve the forecast



Text adds interpretable evidence. It did not improve prediction in this sample.

Mean absolute error in adjusted reviews / \$1M. Ten eligible project outcomes.

Live demo: from a project record to a proposal

1 Inspect the historical evidence

Dilworth outcome, geometry and learned review topics

2 Test a documented scenario

Change the budget, inspect support, then export

Live demonstration: 3 minutes

02 / RESEARCH ESTIMATE

Expected review
activity per dollar

+56.7

growth-adjusted net Yelp reviews / \$1M

Equivalent modeled difference: **+3,120.3 nearby
reviews** across the two-year post period.

Historical error span: -46.1 to +159.6

Historical city holdout error envelope - not a
confidence interval

- The displayed empirical error range uses the maximum absolute city-holdout error, symmetrically around the estimate. It is a historical error envelope, not a confidence or prediction interval and has no guaranteed coverage.
- Project type selects comparable evidence only; changing type does not change the fitted estimate.
- Cost is on the nominal reported historical basis; no causal budget response or inflation conversion is implied.
- The model does not outperform the mean-only comparator in every timing-exclusion sensitivity; its apparent advantage is not robust.

The selected model uses baseline review intensity, reviewed-business count, and cost. Text and trend features are retained as profile context; they did not improve both held-out checks enough to be selected.

49.3 reviews / \$1M

Mean absolute error when each project is held out - City-

The current estimate needs stronger evidence

City-held-out MAE	Model	Mean
All 10 projects	45.6	58.5
Exclude pandemic windows, 8 cases	60.3	59.5
Also exclude construction overlap, 7 cases	70.2	61.6

The predictive advantage does not survive these timing exclusions.

Five geometry proxies remain. Current proposals need fresh local measurements.

A usable foundation for public-project planning

**Review language becomes evidence
that an official can inspect.**

Today: compare historical projects and explore assumptions.

Next: pilot one measured proposal, then validate
with more projects and independently labeled text.

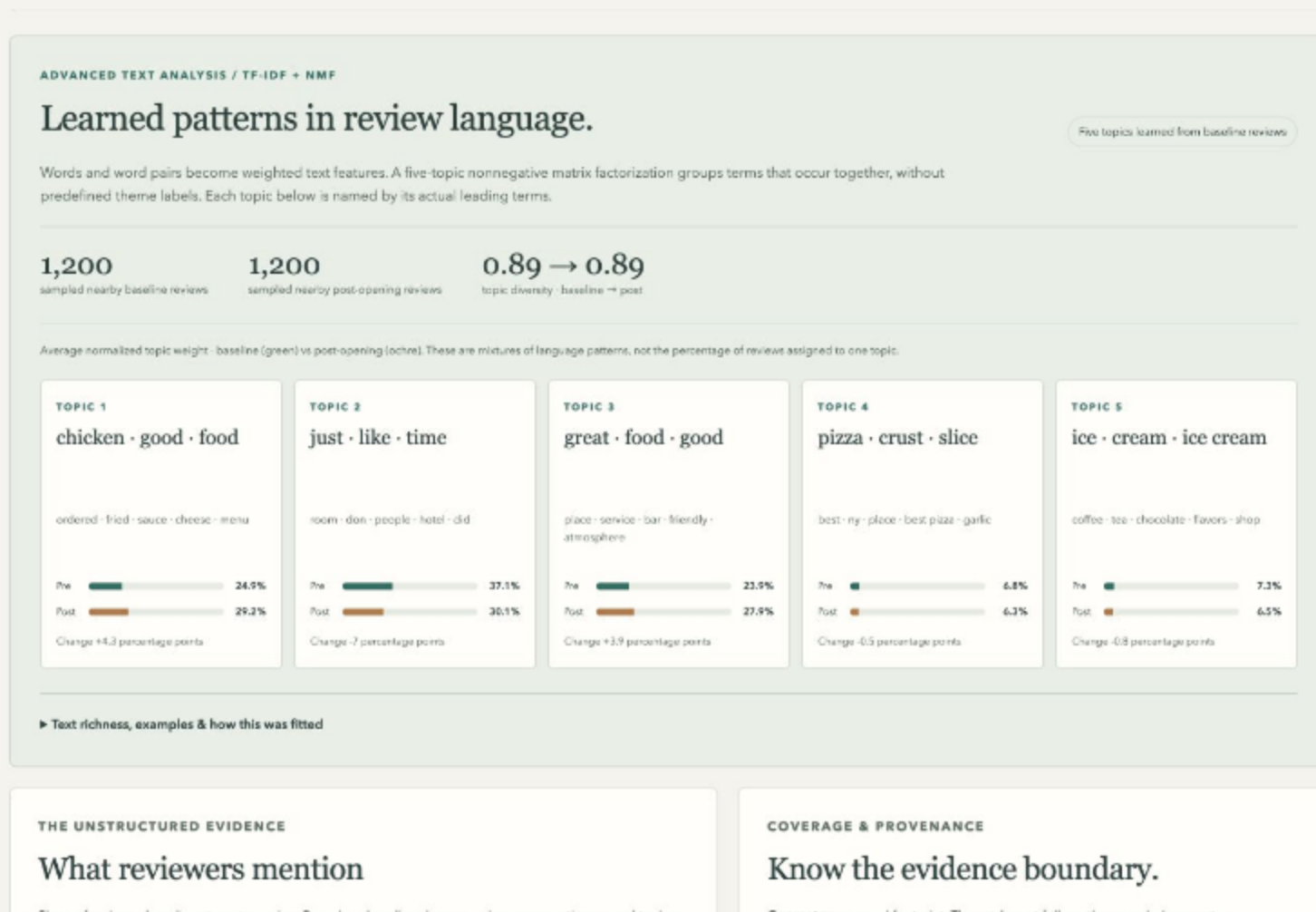
Appendix: model comparisons and inputs

Predictor	Project MAE	City MAE
Mean benchmark	61.62	58.48
Review activity + cost	49.33	45.61
Plus rule text	53.10	48.19
Plus earlier trend	51.52	46.26
Plus learned text	53.04	46.67

Baseline inputs: cost, reviewed businesses and reviews per reviewed business

All errors use adjusted reviews / \$1M. Evaluation remains exploratory.

Appendix: learned-text demo fallback



Appendix: validation and remaining work

Implemented

- Baseline-only topic fitting
- Fold-specific vocabulary and NMF
- Shared-review exclusion
- Versioned profile import and export

Still needed

- Independent human text labels
- Verified historical footprints
- Current, broader project coverage
- Independent forecast validation

The historical error span is empirical, with no calibrated coverage guarantee.